

Mehraz Hossain Rumman

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EXPERIENCE

- **Software Engineer** Dhaka, Bangladesh
Adfinix July 2024 - Present
 - Developed the backend from scratch for **Adsloft**, a SaaS product used to track DSP/SSP clicks, impressions, views, and efficiently generate real-time data and detailed reports for advertisers and publishers.
 - Technologies: **Django, Django Rest Framework, ClickHouse, Next.js, Docker, Jenkins.**
 - Developed a website for Adsloft using **Next.js**, focusing on performance and responsive design.
 - Upgraded the version of another SaaS product, **Adfinix**, from **Django 3.2** to **Django 5.1**, along with other packages to their latest versions. Fixed bugs and optimized code performance for better scalability and efficiency.
 - Fixed API issues and bugs, implemented features such as **CPM** and **frequency capping** in the existing Adfinix product, and developed rich media ads.
 - Worked with **OpenRTB** for real-time bidding processes and developed a mock server to test DSP and SSP request-response interactions.
 - Built the **Aifinix backend from scratch** using **Django** and **Django Rest Framework**, designing scalable APIs, authentication, analytics pipelines, and business logic for an ad-tech platform.
 - Contributed to the **frontend development of Aifinix**, working closely with backend APIs to ensure seamless data flow, performance optimization, and a smooth user experience.
 - Designed and developed an **analytical dashboard end-to-end**, handling **schema design**, backend aggregation logic, and frontend visualization for real-time insights and reporting.
 - Worked extensively with **MongoDB** for storing and querying high-volume analytical and event data, optimizing schemas and queries for performance and scalability.
 - Developed a **Slack bot** for work automation, implementing custom features to streamline internal workflows, notifications, and team productivity.
- **Google Summer of Code** Remote
Contributor May 2025 - Jul 2025 · 3 mos
 - Implemented an advanced HTTP caching feature in the **pyOBIS** Python package to improve performance and efficiency for repeated API calls.
 - Worked on data analysis and visualization for ocean science datasets using Python.
 - Created interactive Jupyter notebooks to demonstrate data workflows and reproducible research.
 - Developed and tested machine learning models for ecological and environmental data.
 - Applied Species Distribution Modeling (SDM) techniques to analyze spatial patterns and biodiversity.
 - Final Report: [Link](#)
 - Organization: **Integrated Ocean Observing System.**
- **Online Python Trainer** Remote
Freelance Nov 2022 - Present · 2 yrs 10 mos
 - Teach students Python programming, from beginner to intermediate levels.
 - Help students debug code, understand errors, and develop strong problem-solving skills.
 - Introduce and explain core concepts in data structures and algorithms.
 - Conduct weekly exams and coding challenges to track progress and reinforce learning.
 - Guide students on practical approaches to solving programming problems.
- **Trainee Software Developer** Dhaka, Bangladesh · Remote
TrendBreaker Apr 2022 - May 2024 · 2 yrs 2 mos
 - Built a desktop application using **PyQt5** for enhanced user interaction and productivity.
 - Developed a **React Native** mobile app and designed the backend using **Django**, delivering a seamless cross-platform experience.
 - Worked on data analysis projects to provide insights and support business decisions.

TECHNICAL SKILLS

- Programming Languages: **Python, Go, C, Fortran, C++, SQL.**
- Databases: **PostgreSQL, Clickhouse, MongoDB.**
- Technologies & Tools: **Docker, CI/CD Pipelines, Git, Github Actions, BitBucket.**
- Familiar with: **Linear Algebra, Matrix, Calculus, Pandas, NumPy, Fuzzy Mathematics, scikit-learn, matplotlib, folium, xarray.**
- Machine Learning Skills: **Supervised Learning (SVM, One-Class SVM, SDM), Unsupervised Learning, Species Distribution Modeling (SDM), Random Forest, Regression,**
- Skills Gained from GSoC: **Data Analysis, Data Preprocessing, Model Development and Testing, Ecological and Environmental Data Analysis, Machine Learning for Biodiversity Analysis, Species Distribution Modeling (SDM).**

COMPETITIVE PROGRAMMING & HACKATHON ACHIEVEMENTS

- **Champion** at, **JnU intra-university programming Contest (solo)** 2021. [Handle: keai33]
- **Champion** at, KYAU National Hackathon. [Team: Quantum Guys]
- **Honourable Mention** at **ICPC 2021**, 11th at UITS IUPC.
- **49th** at **BUET IUPC** [Team: JnU_ABC].
- **46th** at **CoU IUPC** [Team: JnU_Team_Z].
- Became Specialist in **Codeforces** • 3* coder in **CodeChef** • Also solved problems on **Hackerrank** • Solved problems on **LeetCode**.

OPEN SOURCE CONTRIBUTION

- Contributed to **pyOBIS** – code improvements and feature updates.
- Contributed to **Scrapy** – **3 PRs merged**, test fixes, and cleanup.
- Contributed to **Celery** – bug fixes and enhancements.
- Contributed to **DRF-extensions** – **6 PRs merged**, test fixes, and version upgrades.
- Contributed to **Django** – minor improvements and patches.
- Contributed to **Django REST Framework** – fixes and improvements.

PROJECTS

- **KachaMorich: An e commerce app** [Github Organization](#)
Description: Features for clients include account management, purchasing, payments, profile editing, and order reviews. Admin features include user and product management, promoting users, and updating order delivery and payment statuses.
Technologies Used: Django, DjangoRestFramework, React, PostGresql.
- **Digital Signal Processing with Fast Fourier Transform** [Drive Link](#)

EDUCATION

- **Bachelor of Science, Mathematics** Dhaka, Bangladesh
Jagannath University, Dhaka; GPA: 3.59/4.00 Jan 2019 – Feb 2024

REFERENCES

- **Tylar Wayne Cole Murray, PhD.** (ey/em—he/him)
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